

Title

Performance Testing – SB Stocks

Introduction

Performance testing was conducted to evaluate the responsiveness and stability of the SB Stocks application under normal development and user interaction conditions.

Testing Objectives

The primary objectives were:

- Evaluate frontend loading performance.
- Verify API response behavior.
- Test database interaction.
- Verify stock search responsiveness.
- Test portfolio calculations.
- Test trading operation processing.
- Verify application stability.

Test Scenarios

Test 1 – Landing Page Loading

Objective: Verify that the landing page loads successfully.

Expected Result: The page should load without application errors.

Result: Passed.

Test 2 – User Registration

Objective: Verify registration API processing.

Expected Result: A valid account should be created successfully.

Result: Passed.

Test 3 – User Login

Objective: Verify authentication performance.

Expected Result: Valid credentials should return authenticated access.

Result: Passed.

Test 4 – Stock Listing

Objective: Verify stock data retrieval.

Expected Result: Available stocks should be displayed.

Result: Passed.

Test 5 – Stock Search

Objective: Test search response.

Expected Result: Matching stocks should be returned.

Result: Passed.

Test 6 – Portfolio Retrieval

Objective: Verify portfolio API response.

Expected Result: User portfolios should load successfully.

Result: Passed.

Test 7 – Buy Transaction

Objective: Test stock purchase processing.

Expected Result: Balance and portfolio should update correctly.

Result: Passed.

Test 8 – Sell Transaction

Objective: Test stock selling processing.

Expected Result: Holdings and balance should update correctly.

Result: Passed.

Test 9 – Transaction History

Objective: Verify transaction retrieval.

Expected Result: User transactions should be displayed in descending chronological order.

Result: Passed.

Test 10 – Admin Data Retrieval

Objective: Verify admin API access.

Expected Result: Authorized admin users should access administrative information.

Result: Passed.

Validation Testing

The following invalid operations were tested:

- Duplicate email registration.
- Invalid login.
- Missing authentication token.
- Invalid stock quantity.
- Insufficient virtual balance.
- Selling unavailable stocks.
- Selling excess stock quantity.
- Unauthorized portfolio access.
- Unauthorized admin access.

The application rejected invalid operations with appropriate error responses.

Performance Observation

The application demonstrated acceptable responsiveness under normal project testing conditions.

Frontend navigation remained responsive, and API requests completed within reasonable development-environment response times.

MongoDB successfully handled user, portfolio, transaction, and order operations.

Conclusion

Performance testing indicates that SB Stocks provides stable and responsive operation for its intended educational and paper trading use case.